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RESEARCH ARTICLE

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Leveraging machine learning to predict de novo skin malignancy following lung transplantation

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ABSTRACT

Aims: This study aimed to predict post-transplant malignancy risks at multiple levels among lung transplant recipients using machine learning (ML) and to identify key clinical and immunogenetic predictors.

Materials and methods: A dataset of 30,917 lung transplant recipients with no prior cancer history was analyzed using pre-, peri-, and post-transplant variables. Multiple ML algorithms—gradient boosting, random forest, neural networks, and logistic regression—were applied to predict: (1) overall de novo malignancies (DNM), (2) skin versus non-skin cancers, and (3) skin cancer subtypes, including basal cell carcinoma (BCC) and squamous cell carcinoma (SCC).

Results: Gradient boosting achieved the highest AUC for overall malignancies (0.746) and skin versus non-skin cancers (0.642), while random forest performed best for BCC versus SCC classification (AUC = 0.726). Significant predictors included HLA-DR alleles (DR52, DR1, DR53), A locus mismatch, recipient ethnicity, BMI, serum albumin, CMV/EBV serostatus, and cardiac-related measures (LV remodeling, cardiac output, prior cardiac surgery). Additional subtype predictors included peak PRA Class I sensitization, insulin signaling, donor-derived transfusions, and waiting list duration.

Conclusions: ML-driven predictive modeling enables personalized assessment of post-transplant malignancy risk, supporting early detection, targeted surveillance, and optimized long-term care for lung transplant recipients.

ARTICLE HIGHLIGHTS

- Machine Learning Models Predict De Novo Malignancies in Lung Transplant Recipients: using a national cohort of 30,917 lung transplant recipients, gradient boosting achieved the highest AUC for overall malignancy prediction, as well as skin vs. non-skin cancers; random forest excelled in distinguishing classified basal cell carcinoma (BCC) versus squamous cell carcinoma (SCC).
- Key Immunogenetic Predictors Identified: HLA-DR52 linked to reduced overall malignancy risk; HLA-DR1 and A locus mismatch associated with higher skin cancer risk; HLA-DR53 tied to increased SCC over BCC likelihood.
- Clinical and Demographic Risk Factors: Recipient ethnicity (white highest for skin cancers), EBV serostatus, CMV IgG positivity, BMI, serum albumin, age, male gender, and cardiac measures (e.g., output, LV remodeling, prior surgery) emerged as significant across malignancy levels.
- Skin Cancer Subtype-Specific Insights: Low peak PRA Class I sensitization, donor-derived transfusions, lower serum albumin, and shorter waiting list duration favored SCC over BCC; donor non-insulin use reduced SCC odds relative to BCC.
- Potential for Personalized Surveillance: Models enable risk stratification using pre-, peri-, and post-transplant variables, supporting targeted screening and optimized immunosuppression to improve long-term outcomes.

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Lung transplantation; De novo malignancy; machine learning; predictive modeling; HLA-DR; skin cancer

1. Introduction

For individuals with end-stage organ failure, transplantation offers a critical, potentially life-saving treatment. Advances in post-transplant care have significantly improved survival rates within the first year following transplantation. However, despite these advancements in solid organ transplantation outcomes, complications—including an

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elevated risk of malignancy—persist as major concerns. Immunosuppressive medications, while essential for preventing organ rejection, compromise the immune system's ability to detect and eliminate malignant cells, thereby increasing susceptibility to cancer development [1].

Transplant recipients face a substantially increased risk of developing de novo malignancies (DNM), with incidence rates reported to be two to seven times higher than those observed in the general population [2]. Non-melanoma skin cancers, particularly squamous cell carcinoma (SCC) and basal cell carcinoma (BCC), are the most common, constituting a significant proportion of cases [3]. Other prevalent malignancies include lung cancer, lymphoproliferative disorders, colorectal cancer, breast cancer, Kaposi sarcoma, hepatic cancers, and genital cancers [4]. Given the heightened and varied cancer risk in this population, effective surveillance and early detection strategies are essential to improving long-term outcomes in transplant recipients.

Recent progress in explainable artificial intelligence (AI) and machine learning (ML) offers promising opportunities to address this challenge by enhancing risk prediction and early detection of post-transplant malignancies. The increasing availability of big data, including comprehensive datasets from the United States Organ Procurement and Transplantation Network for Organ Sharing (UNOS) transplant registry, has facilitated the development of advanced predictive models [5]. These ML-driven approaches excel at uncovering intricate patterns and subtle risk factors that conventional statistical methods may overlook, enabling more tailored and preventive surveillance strategies. Notably, such algorithms have already proven effective in predicting a broad spectrum of clinical outcomes in solid organ transplant recipients, underscoring their transformative potential for improving post-transplant care and long-term patient survival [6–8].

Prior research suggests that ML algorithms hold considerable promise in predicting graft and patient survival by leveraging pre-transplant data such as waiting lists [6–8]. However, most existing studies have primarily focused on predicting overall survival rates [9–11] rather than identifying prognostic factors specifically associated with post-transplant malignancies. Only a limited number of studies have explored the prediction of de novo malignancy outcomes post-organ transplant, using longitudinal data collected both before and after transplantation [12,13]. However, none of these studies have specifically utilized malignancy data following organ transplant and addressed de novo malignancy of different types. Therefore, the objective of this research is to employ a machine learning approach to construct a series of predictive models for predicting de novo malignancies in patients who have undergone lung transplantation, as well as predicting the types of malignancies patients develop following lung transplantation. Given the prevalence of skin cancers among de novo malignancies following organ transplants [3], we built additional models to classify the type of skin cancers patients develop following lung transplantation. Utilizing data obtained from the Standard Transplant Analysis and Research (STAR), we developed multiple predictive models to classify de novo malignancy patterns and their types in lung transplant recipients and identify significant prognostic factors impacting their risk of developing malignancy skin post-transplantation. The medical records of lung transplant recipients between 1984 and 2021 were utilized for this study. A training set was used for developing prediction models, while a validation set was created for model validation to ensure robustness and generalizability to unseen data. All algorithms were tested on the STAR database, and the best-performing model for our study was reported. The performance of these models was evaluated using the area under the receiver operating characteristic (ROC) curve, along with other performance metrics. Importantly, the study incorporated a range of clinically relevant predictors—including immunosuppression regimens, donor characteristics, and patient demographics—to enhance the clinical applicability of the models.

2. Literature review

Organ transplantation remains one of the most effective therapeutic interventions for patients with end-stage respiratory failure [14]. With the rapid integration of artificial intelligence and machine learning into clinical practice, transplant clinicians are increasingly leveraging these technologies as powerful decision-support tools. Applications include predicting waiting list mortality, optimizing donor selection, forecasting patient and graft survival rates, predicting de novo malignancies, and refining immunosuppression strategies to minimize the risk of rejection [8,15].

Machine learning plays an increasingly critical role in addressing the challenges associated with prolonged waiting times for organ transplantation, which raise significant concerns regarding equitable access to donor organs [8]. Accurate prediction of waiting list mortality and morbidity is essential for making informed decisions about patient listing, organ allocation, and donor organ acceptance. ML algorithms have demonstrated strong performance in uncovering previously unrecognized risk factors that influence transplant outcomes by integrating diverse variables such as patient demographics, genetic profiles, comorbidities, and graft quality [16].

Recent research has increasingly focused on optimizing post-transplant outcomes by incorporating both donor and recipient characteristics to enhance transplant success rates. The complexity and ethical considerations of organ allocation and donor-recipient matching make real-world testing of alternative allocation systems challenging. However, ML can provide an alternative framework for simulating and evaluating diverse allocation scenarios prior to clinical implementation. For example, Davis, Mehrotra [17] utilized multiperiod linear optimization models to evaluate the potential impact of various kidney-sharing strategies before policy adoption. Similarly, other studies have applied ML approaches, including artificial neural networks (ANN), to donor and recipient datasets to explore a range of organ matching methodologies for liver transplantation [18,19].

Given the persistent shortage of donor organs relative to patient demand, accurate prediction of organ transplantation outcomes is essential for optimizing patient care and resource allocation [8]. Multiple studies have developed machine learning algorithms to identify key predictive factors influencing post-transplant survival [20–23]. These models have pinpointed critical factors such as graft failure, infections, renal failure, and vascular complications. Additionally, ML has been effectively utilized to predict short-term postoperative survival in organ transplantation by leveraging preoperative physiological data, including blood test results collected 1–9 days before surgery [24]. Similarly, Lau, Kankanige [25] developed a personalized ML-based predictive model for 90-day graft failure in pediatric liver transplantation, identifying key predictors such as preoperative hepatic encephalopathy, sodium levels at the end of surgery, hepatic artery thrombosis, and total bilirubin levels.

In thoracic transplantation, Amini, Bagheri [26] developed an analytical framework to identify key determinants of long-term survival following lung transplantation. Using data from over 44,000 lung transplant recipients, they identified factors such as hepatitis B surface antibody levels and forced expiratory volume in one second (FEV1) as significant predictors, providing valuable insights for optimizing lung allocation strategies. In a related study, Medved, Ohlsson [27] compared a deep-learning-based prediction model with a conventional risk-based model to predict short- and long-term survival after heart transplantation. Their results demonstrated that the deep-learning model outperformed the traditional approach, offering superior accuracy in forecasting survival outcomes.

Despite several studies addressing various aspects of organ transplantation, limited attention has been devoted to evaluating the risk and types of de novo malignancies specifically following lung transplantation. To bridge this critical gap, we extend our previous study [12] by developing machine learning models specifically designed to predict both the occurrence and classification of de novo malignancies, including skin cancer subtypes, in lung transplant recipients. By leveraging longitudinal datasets that integrate both pre- and post-transplant variables, these models aim to enhance individualized risk assessment and support more effective, targeted post-transplant surveillance strategies.

3. Materials and methods

Our research methodology consists of three primary components: data acquisition, data preparation, and data modeling [28]. Each of these elements is described in detail in the following subsections and summarized through a comprehensive visualization in Figure 1. The data set used in this study was obtained from the Standard Transplant Analysis and Research (STAR) data sources, which are administered by the United Network for Organ Sharing (UNOS). STAR supports the operations of the Organ Procurement and Transplantation Network (OPTN) in the United States and contains detailed, individual-level information on all organ transplants performed nationwide from 1987 to the present. As the most comprehensive organ transplantation database in the US, STAR provides the necessary longitudinal depth and breadth required for predictive modeling. To prepare the data for analysis, extensive data cleaning and preprocessing were conducted to ensure quality and consistency, followed by transformation into a suitable format for modeling. SAS Enterprise Miner was employed for data analysis, exploration, and preparation of the predictive modeling framework.

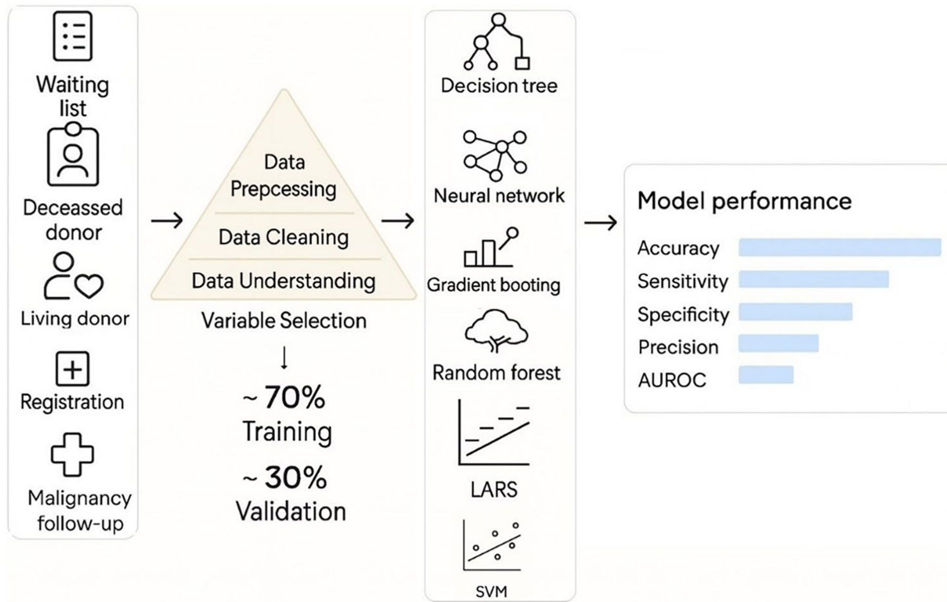

Figure 1. Research framework.

3.1. Data acquisitions

In our study, we compiled a comprehensive dataset by integrating information from multiple databases including the Waiting List Data (WLD), Malignancy Follow-up (FOL), Recipient HLA (RH), Transplant Candidate Registration (TCR), Deceased Donor Registration (DDR), Living Donor Registration (LDR), Transplant Recipient Registration (TRR), as illustrated in Figure 1.

The waiting list database provided essential insights into patient demographics, panel-reactive antibody (PRA) sensitivity, human leukocyte antigen (HLA) profiles, and other baseline clinical characteristics. The Transplant Candidate Registration (TCR) and Transplant Recipient Registration (TRR) databases included detailed clinical, physiological, and procedural data collected from the time of listing through transplantation. The Recipient HLA (RH) database contained immunologic profiles, including detailed HLA typing results, relevant for graft compatibility and immune response. The Deceased Donor Registration (DDR) and Living Donor Registration (LDR) databases provided donor-related information such as demographics, serology results, organ recovery methods, and preservation techniques. The Malignancy Follow-up (FOL) database contained longitudinal information on post-transplant malignancies, including cancer diagnoses and outcomes. After excluding simultaneous heart-lung transplantations, the resultant dataset, exclusively focused on lung transplantation, comprised 66,590 patient records representing individuals who underwent lung transplantation.

3.2. Data preparation

Following the integration of data from multiple sources, significant effort was devoted to preprocessing, cleaning, and analyzing the dataset. To specifically investigate de novo malignancies occurring after lung transplantation, we identified patients diagnosed with malignant cancers during their follow-up periods. Individuals with a documented history of malignancy before transplantation, as noted in their transplant candidate forms, were excluded. This resulted in a dataset of 38,788 cases with no prior malignancy history. After merging this dataset with lung malignancy follow-up data and excluding individuals lost to follow-up, the resultant dataset comprised 30,917 individuals, of whom approximately 25% (~7,566 cases) developed de novo malignancies post-lung transplantation.

Following the merging process, the compiled dataset contained a total of 580 variables. To streamline the analysis and focus on the most relevant predictors, each variable was carefully evaluated based on several criteria, including clinical relevance, redundancy, and the presence of extensive missing values (>30%). Additionally, the strength of association between each predictor and the target variable was assessed to

Table 1. Baseline characteristics of lung transplant recipients with and without de novo malignancies.

Variable	Total cohort (N = 30,917)	No DNM (n = 23,351)	DNM (n = 7,566)	p-Value
Age (years)*	52.82 ± 15.00	51.85 ± 15.89	56.43 ± 11.90	<0.001
Weight (kg)*	71.45 ± 18.70	70.34 ± 18.90	74.83 ± 17.69	<0.001
Sex				<0.001
Male	17,589 (56.9%)	12,478 (53.6%)	5,051 (66.1%)	
Female	13,338 (43.1%)	10,797 (46.4%)	2,591 (33.9%)	
Ethnicity				<0.0001
White	25,578 (82.7%)	18,540 (79.6%)	7,038 (92.1%)	
Non-White	5,339 (17.3%)	4,735 (20.4%)	604 (7.9%)	
Cigarette smoking history				<0.0001
Yes	13,733 (57.1%)	10,043 (54.4%)	3,690 (68.1%)	
No	10,302 (42.9%)	8,415 (45.6%)	1,887 (31.9%)	
Primary diagnosis				<0.0001
Idiopathic pulmonary fibrosis & ILD	10,721 (34.7%)	7,934 (34.0%)	2,787 (36.8%)	
COPD/Alpha-1 antitrypsin deficiency	9,394 (30.4%)	6,702 (28.7%)	2,692 (35.6%)	
Cystic fibrosis	4,276 (13.8%)	3,617 (15.5%)	659 (8.7%)	
Pulmonary hypertension	797 (2.6%)	674 (2.9%)	123 (1.6%)	
Sarcoidosis/Hypersensitivity	2,246 (7.3%)	1,700 (7.3%)	546 (7.2%)	
Other/Unknown	3,483 (11.3%)	2,724 (11.7%)	759 (10.0%)	
Prior HBV infection				<0.0001
Negative	26,242 (87.47%)	19,920 (66.40%)	6,322 (21.07%)	
Positive	1,148 (3.83%)	908 (3.03%)	240 (0.80%)	
Not tested/Unknown	2,610 (8.70%)	1,806 (6.01%)	804 (2.69%)	
HCV infection				<0.0001
Negative	28,390 (92.7%)	21,488 (93.1%)	6,922 (91.7%)	
Positive	576 (1.88%)	448 (1.94%)	128 (1.7%)	
Not tested/Unknown	1,648 (5.39%)	1,145 (4.96%)	503 (6.72%)	
CMV IgG positive				<0.001
Negative	8,824 (51.11%)	6,804 (51.61%)	3,020 (49.88%)	
Positive	870 (4.53%)	780 (5.91%)	90 (1.49%)	
Not tested/Unknown	7,719 (44.36%)	5,732 (43.48%)	1,987 (48.63%)	
EBV seropositive				<0.001
Negative	22,860 (80.50%)	17,542 (81.29%)	5,318 (78.03%)	
Positive	3,208 (11.24%)	2,405 (11.14%)	803 (11.55%)	
Not Tested/Unknown	2,454 (8.60%)	1,633 (7.57%)	821 (10.42%)	
Total serum albumin (g/dL)*	3.89 ± 0.63	3.86 ± 0.65	3.97 ± 0.60	<0.001
Follow-up time (days)*	1852.72 ± 1568.98	1551.05 ± 1381.71	2771.51 ± 1738.85	<0.001

*Mean ± SD.

identify and eliminate highly correlated variables that could negatively impact model performance. As a result of this comprehensive evaluation, 173 variables were excluded from further analysis, leaving a refined set of 407 variables, including detailed information on follow-up malignancy subtypes, for subsequent predictive modeling.

Based on the compiled dataset, key variables were categorized into three distinct groups: (1) pre-transplant measures, including recipient and donor demographic data such as gender, race, and age; (2) during-transplant measures, encompassing blood work results; and (3) post-transplant measures, covering patient outcomes such as survival, hospital length of stay, occurrence of malignancy, cancer types and other complications. Table 1 summarizes the baseline demographic, clinical, and transplant-related characteristics of the study cohort, comparing recipients who developed de novo malignancies with those who did not, highlighting significant differences in factors such as age, sex, ethnicity, primary diagnosis, and key viral serostatus markers.

Table 2 provides a detailed summary of the different types of de novo malignancies observed in lung transplant recipients, along with their corresponding case counts and percentage distributions. As the data indicate, skin cancers—including skin SCC, BCC, and melanoma—dominate, accounting for 54.8% of all cases. Lymphoid malignancies (e.g., lymphoma and leukemia) represent 10.5%, while thoracic cancers, primarily lung cancer, contribute 8.7%. Digestive cancers account for 6.9%, and genital cancers comprise 4.7%. The remaining cases are distributed among other categories, including central nervous system (CNS) tumors, breast cancers, head and neck cancers, sarcomas, and endocrine cancers, each representing smaller proportions of the total malignancy cases. As shown in Figure 2, among skin-related cancers, SCC is the most common post-transplant malignancy, accounting for approximately 44% of cases. Basal cell carcinoma (BCC) is also relatively frequent, representing about 11% of cases, while melanoma is rare, occurring in only about 1% of cases. Figure 3 presents the demographic distribution of the skin cancer cases included in this study.

Table 2. Summary of De novo malignancies and their prevalence in lung transplants.

Cancer type	Frequency	Percentage (%)
Skin squamous cell carcinoma	3628	43.74
Skin basal cell carcinoma	878	10.58
Malignant de novo lymphoma	839	10.11
Lung cancer	737	8.88
Other cancers	471	5.68
Colorectal cancer	223	2.69
Brain tumor	197	2.37
Prostate cancer	155	1.87
Bladder cancer	150	1.81
Breast cancer	122	1.47
Liver metastasis	114	1.37
Skin melanoma	112	1.35
Renal carcinoma	79	0.95
Pancreatic cancer	78	0.94
Esophageal cancer	58	0.70
Leukemia	50	0.60
Primary hepatic cancer	49	0.59
Vulvar, penile, scrotal carcinoma	49	0.59
Primary unknown cancer	46	0.55
Tongue and throat cancer	43	0.52
Stomach cancer	40	0.48
Sarcoma	34	0.41
Laryngeal cancer	33	0.40
Thyroid cancer	29	0.35
Ovarian cancer	21	0.25
Small intestine cancer	20	0.24
Uterine carcinoma	15	0.18
Kaposi's sarcoma (cutaneous)	13	0.16
Kaposi's sarcoma (visceral)	9	0.11
Testicular cancer	3	0.04

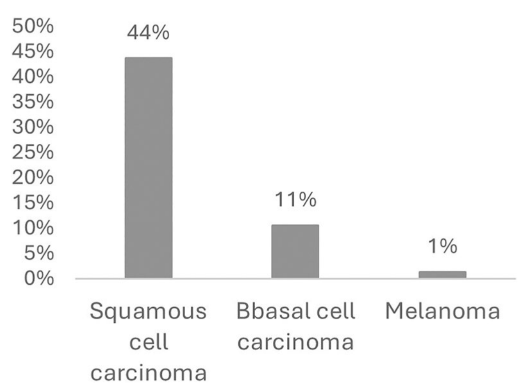

Figure 2. Skin cancer types distribution.

3.3. Data modelling

This study conducted a supervised machine learning investigation to gain deeper insights into the occurrence of malignancies and the types of skin malignancies among lung transplant recipients. To achieve this objective, three distinct classes of predictive models were developed: decision trees, logistic regression, and neural networks. The decision tree models included entropy-based trees, boosting trees, Gini index, chi-square, and random forests. Logistic regression modeling incorporated multiple approaches, including stepwise, backward, forward, and least angle regression techniques. Each model, whether a decision tree, logistic regression, or neural network, was developed following industry best practices to ensure methodological rigor and robust empirical analysis of malignancy patterns. For model evaluation, the dataset was randomly divided into training (70%) and testing (30%) subsets. To address class imbalance between malignancy and non-malignancy outcomes, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data. This method generates synthetic samples of the minority class rather than simply duplicating existing ones, improving model generalizability and reducing overfitting.

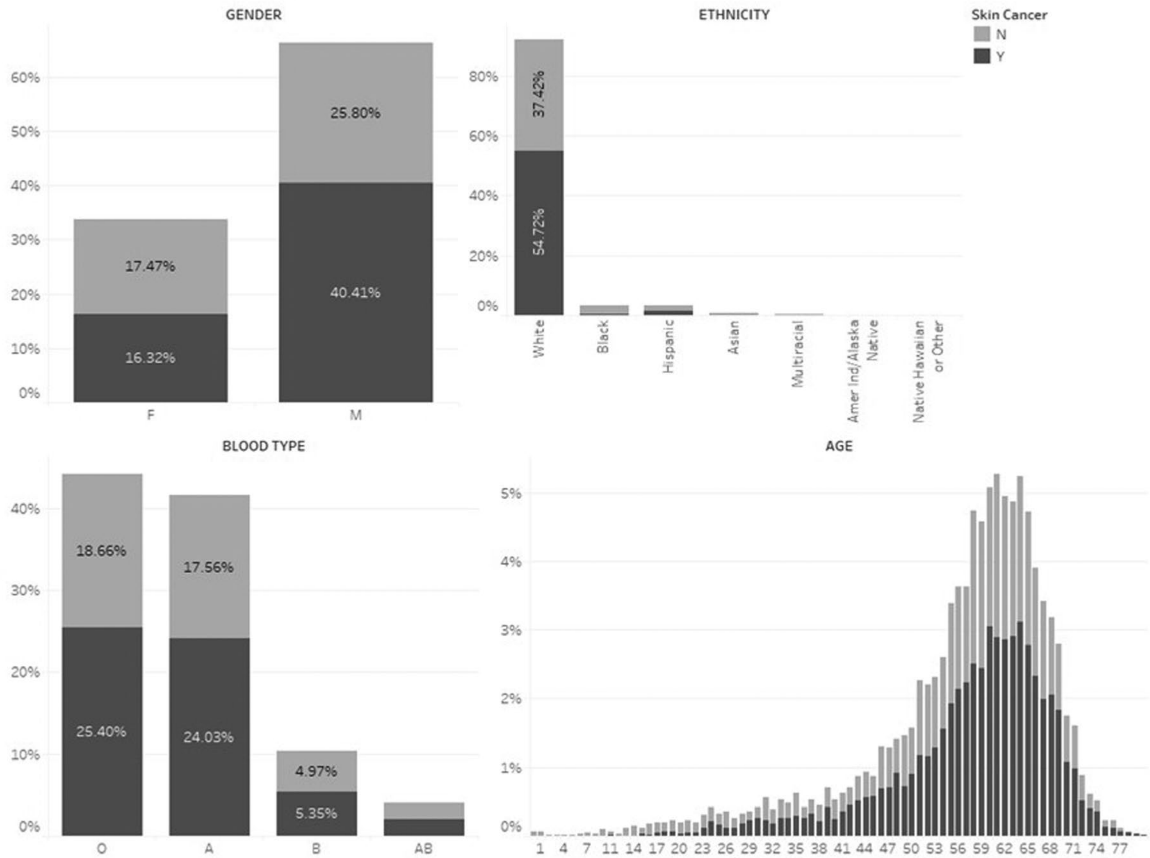

Figure 3. Demographic summary of skin cancer malignancies by gender, ethnicity, blood type, and age.

3.3.1. Logistic regression

Logistic regression is a widely used machine learning technique for predicting the probability of an outcome based on one or more predictor variables [29]. In this study, logistic regression models were developed to predict the presence or absence of malignant cancers in lung transplant recipients, where the target variable was binary (malignancy = 1, no malignancy = 0). Additional models were also built to predict the occurrence of skin cancers and to classify skin cancer subtypes, given their high prevalence among post-transplant malignancies. To identify the most influential predictors, three variable selection methods—forward selection, backward elimination, and stepwise selection—were applied [30]. Among these, the stepwise selection method consistently produced the most optimal models across all three classification tasks, balancing accuracy and interpretability by retaining only the most significant predictors of malignancy risk.

3.3.2. Least angle regressions (LARS)

The Least Angle Regression (LARS) algorithm is a machine learning technique designed to efficiently handle high-dimensional data and multicollinearity by selecting only the most relevant predictors. Unlike standard regression models, LARS uses model selection criteria such as Schwartz's Bayesian Criterion (SBC), Akaike's Information Criterion (AIC), or adjusted R^2 to identify optimal predictors. It incrementally builds the model by adding variables based on their correlation with the target outcome, making it particularly effective for datasets with many potential predictors. In this study, SBC was chosen as the selection criterion to ensure the most optimal LARS model [31,32].

3.3.3. Decision tree

Decision trees are widely used for their ability to handle missing data and provide easily interpretable results. They work by recursively splitting data into branches based on input variables, progressively improving classification accuracy [33,34]. In this study, several decision tree algorithms were evaluated to identify the optimal leaf-node splitting criteria, including the Chi-square log-worth, Gini index, and Entropy-based information gain methods [35,36]. These algorithms assess variable significance and determine the best thresholds for splitting the data. To avoid overfitting, pruning techniques were applied to simplify the trees and enhance generalizability. Among the evaluated methods, the Chi-square log-worth decision tree achieved the best overall performance, demonstrating superior accuracy, precision, and AUROC across all three classification tasks.

3.3.4. Gradient boosting

Gradient boosting is an ensemble learning technique that builds a sequence of decision trees, with each tree trained to correct the residual errors of the previous model, thereby progressively improving prediction accuracy [37,38]. In this study, gradient boosting was applied to identify important predictors that were not highlighted by other decision tree algorithms and to enhance overall model performance. Several key parameters were tuned to optimize the model, including the number of iterations, the proportion of observations used for training each tree, and the shrinkage factor (learning rate), which was adjusted to minimize the risk of overfitting [39]. This configuration allowed the gradient boosting model to achieve strong predictive performance while maintaining model generalizability.

3.3.5. Random Forest

Random forest is an ensemble learning method that combines the predictions of multiple decision trees to generate more accurate and robust estimates of the target variable [40]. It uses bootstrap sampling to create random subsets of the training data, builds unpruned trees on each subset, and aggregates their predictions by majority voting to determine the final class label [41]. In this study, the random forest model was optimized by tuning three key parameters: the maximum number of trees, the proportion of training data used for each tree, and the number of input variables considered at each split. These parameter settings were chosen to enhance each model's predictive accuracy and generalizability, particularly in handling high-dimensional data and missing values, where random forests typically outperform standard decision trees.

3.3.6. Neural networks

Artificial neural networks (ANNs) are a class of machine learning algorithms capable of capturing complex, nonlinear relationships among variables, inspired by the functioning of the human brain [42]. In this study, we employed a Multilayer Perceptron (MLP), a widely used feed-forward neural network architecture, consisting of an input layer representing predictor variables, one or more hidden layers that process nonlinear interactions, and an output layer representing the target variable. The model learns by assigning and iteratively adjusting connection weights between neurons during training to minimize prediction errors. Activation functions, such as logistic or hyperbolic tangent functions, were used to introduce nonlinearity, allowing the network to effectively capture intricate patterns in the data. The MLP was trained using a conjugate-gradient optimization method to improve predictive performance while minimizing error.

3.3.7. Support vector machine

Support Vector Machines (SVMs) are a powerful class of machine learning algorithms used for classification, outlier detection, and regression tasks [43]. SVMs transform input data into a higher-dimensional space using nonlinear kernels such as polynomial and sigmoid functions, enabling the algorithm to create an optimal separating boundary between classes. The primary objective is to maximize the margin between these separating hyperplanes, thereby improving classification accuracy. SVMs are particularly advantageous for handling imbalanced datasets and offer lower computational complexity compared to more resource-intensive models, such as artificial neural networks.

Supplementary Appendix A summarizes the key hyperparameters configured for each machine learning model to ensure optimal performance and comparability across predictive analyses.

3.3.8. Measures for performance evaluation

In this study, model performance was evaluated using four commonly used binary classification metrics. Accuracy measures the overall correctness of the model's predictions. Misclassification rate represents the proportion of incorrect predictions, calculated as the ratio of false positives and false negatives to the total number of cases. Sensitivity (also known as recall) assesses the model's ability to correctly identify positive cases, while specificity measures its ability to correctly identify negative cases. Together, these metrics provide a comprehensive assessment of each model's predictive performance.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Sensitivity = \frac{TP}{TP + FN}$$

$$Specificity = \frac{TN}{TN + FP}$$

$$Specificity = \frac{TN}{TN + FP}$$

$$Precision = \frac{TP}{TP + FP}$$

Here, *TP* represents the number of true positives, *TN* the number of true negatives, *FP* the number of false positives, and *FN* the number of false negatives. These metrics were used to evaluate the performance of the predictive models. Accuracy serves as a comprehensive indicator of model performance, while sensitivity was prioritized due to its critical role in correctly identifying patients with malignancies and accurately classifying their types after lung transplantation.

4. Results

Supplementary Appendix B summarizes the performance of the predictive models, reporting the misclassification rate, accuracy, sensitivity, and specificity for each classifier. Among the models, the decision tree achieved the highest overall classification accuracy. At the same time, the random forest demonstrated the highest sensitivity, effectively identifying a greater proportion of true positive cases compared to the other models. Additionally, model performance was further evaluated using the receiver operating characteristic (ROC) analysis. As illustrated in Supplementary Appendix B, the gradient boosting model outperformed all other classifiers, achieving the highest ROC value of 0.74, indicating superior discriminative ability.

Table 3 presents the most significant variables for predicting post-transplant malignancy, as identified by the gradient boosting model. The assigned values represent the relative importance of each variable, with higher values indicating greater predictive significance.

As noted earlier, skin cancers account for approximately 55% of all de novo malignancy cases following lung transplantation, making them the most common post-transplant malignancies. Among these, squamous cell carcinoma (SCC) is the most prevalent and often aggressive, while basal cell carcinoma (BCC) is less aggressive but still requires close monitoring. Melanoma, though rarer, is highly dangerous, underscoring the importance of early detection. Despite this burden, few studies have developed predictive models to distinguish skin cancers from other malignancies in lung transplant recipients. To address this gap, we built a series of machine learning models to classify skin-related malignancies (SCC, BCC, and melanoma) versus non-skin malignancies among lung transplant patients, recognizing their elevated risk due to long-term immunosuppression. Patients who developed both skin and non-skin cancers were excluded from the analysis, as they represented a small subset and could introduce classification ambiguity.

Supplementary Appendix C summarizes the performance of the predictive models. Among the evaluated approaches, the gradient boosting model achieved the highest overall performance, with an AUROC of 0.642, demonstrating its effectiveness in distinguishing skin versus non-skin malignancies in lung transplant recipients.

Table 3. Key variables identified by gradient boosting model for de novo malignancy prediction after lung transplantation.

Variable	Description	From	Validation importance	Rank
DR52	Candidate Most Recent/at Removal DR52 Antigen	WLD	1.34	1
END_BMI_CALC	Calculated Candidate BMI	CALCULATED	1.33	2
CMV_IGG	Person was infected with CMV	TRR	1.29	3
TOT_SERUM_ALBUM	Recipient total serum albumin at registration	TCR	1.16	4
ECMO_72HOURS	Most recent Recipient's international normalized ratio (INR)	TRR	1.14	5
EBV_SEROSTATUS	Serological status of an individual regarding Epstein-Barr virus (EBV) infection	CALCULATED	1.07	6
LOS	Recipient length of stay	TRR-CALCULATED	1.03	7
TRTREJ1Y	Calculated-treated for rejection within 1 year	CALCULATED	0.97	8
ETHCAT	Recipient ethnicity category	TCR	0.87	9
GENDER	Recipient gender	TCR	0.82	10
INIT_AGE	Age in years at time of listing	CALCULATED	0.80	11
BUN_DON	Deceased donor-terminal blood urea nitrogen	CALCULATED	0.68	12

Table 4. Key variables identified by gradient boosting model for skin vs. non-skin malignancy prediction after lung transplantation.

Variable	Description	From	Validation importance	Rank
ETHCAT	Recipient ethnicity category	TCR	1.00	1
EBV_SEROSTATUS	Recipient EBV status at transplant	CALCULATED	0.59	2
AGE	Recipient age (yrs)	TRR-CALCULATED	0.53	3
HEMO_CO_TCR	Most recent hemodynamics CO L/min at registration	TCR	0.32	4
GENDER	Recipient gender	TCR	0.32	5
HGT_CM_CALC	Calculated recipient height (cm)	CALCULATED	0.30	6
AMIS	HLA mismatch level	CALCULATED	0.29	7
TRTREJ1Y	Treated for rejection within 1 year	CALCULATED	0.21	8
PRIOR_CARD_SURG_TCR	TCR prior cardiac surgery at listing (non-transplant)	TCR	0.15	9
RDR1	Recipient DR1 antigen	RH	0.15	10
HEMO_PA_MN_TCR	Most recent hemodynamics PA (mean) mm/Hg at registration	TCR	0.09	11
LOS	Recipient length of stay post TX	TRR-CALCULATED	0.09	12
DAYSWAIT_CHRON	Total days on waiting list	CALCULATED	0.08	13
TCR_DUR_ABSTAIN	Duration of abstinence for cigarette use	TCR	0.05	14
RDR2	Recipient DR2 antigen	RH	0.04	15

Table 4 presents the top predictive variables identified by the gradient boosting model for distinguishing between skin and non-skin malignancies. The reported validation importance scores reflect each variable's contribution to the model's predictive performance, where higher values indicate greater influence.

We then restricted our dataset to patients who developed any form of skin cancer and focused on predicting skin-based malignancies. Within this group, we concentrated on non-melanoma skin cancers (NMSC)—specifically basal cell carcinoma (BCC) and squamous cell carcinoma (SCC)—as they represent the most common malignancies in transplant recipients. In contrast, melanoma cases were excluded from this analysis because they are rare, as shown in Table 2, accounting for only ~1% of all skin cancers in this population.

Supplementary Appendix D summarizes the performance of several machine learning models developed to classify BCC versus SCC among lung transplant recipients. Among all evaluated models, the random forest model achieved the highest predictive performance with an accuracy of 82.6% and an AUROC of 0.726, demonstrating strong predictive ability. The boosting and decision tree models also performed competitively, whereas logistic regression and rule induction showed comparatively lower predictive performance.

Table 5 lists the most influential variables identified by the random forest model for differentiating BCC from SCC. The reported validation importance scores are based on Gini reduction, where higher scores indicate a greater contribution to the model's predictive performance.

Table 5. Key variables identified by random Forest model for BCC vs. SCC malignancy prediction after lung transplantation.

Variable	Description	From	Validation importance	Rank
PRAPK_CL1	Recipient peak PRA% class I at transplant	RH	0.66	1
TRANSFUS_INTRAOP__DON	Archived DDR transfusion units intraoperatively	DRR	0.64	2
TOT_SERUM_ALBUM	Patient total serum albumin at registration (pre 1/1/2007 for adult)	TCR	0.54	3
LEFT_VENT_REMODEL	Left ventricular remodeling occurring before listing	TCR	0.52	4
HIST_INSULIN_DEP_DON	Deceased donor-insulin dependent diabetes (Y,N)	DDR	0.48	5
DR53	Candidate most recent/at removal DR53 antigen from waiting list	WLD	0.43	6
DATA_WAITLIST	Candidate WL data reported	CALCULATED	0.38	7
PULM_INF_CONF_DON	Deceased donor-infection pulmonary source-confirmed	DDR	0.31	8
HEMO_CO_TCR	Most recent hemodynamics CO L/min at registration	TCR	0.26	9
HEMO_PCW_TCR	Most recent hemodynamics PCW (mean) mm/Hg at registration	TCR	0.26	10
ABO_MAT	Donor-recipient ABO match level	CALCULATED	0.22	11
HIST_OTH_DRUG_DON	Deceased donor-history of other drug use in past	DDR	0.20	12
DQ1	Candidate most recent/at removal DQB1 antigen from waiting list	WLD	0.19	13
HBV_SUR_ANTIGEN	Recipient hep B surface antigen	TRR	0.11	14
DDR1	Donor DR1 antigen	CALCULATED	0.09	15
HBV_CORE	Recipient hepatitis B-core antibody	TRR	0.07	16
DIAB	Recipient diabetes at registration	TCR	0.06	17
DA2	Donor A2 antigen	CALCULATED	0.05	18
RESIST_INF	Pan-resistant bacterial infection at registration	TCR	0.04	19
DB1	Donor B1 antigen	CALCULATED	0.01	20

5. Discussion

Advances in lung transplantation have significantly improved patient survival; however, post-transplant malignancies remain a major cause of long-term morbidity and mortality. Among these, skin malignancies represent a growing clinical challenge, largely driven by lifelong immunosuppression and complex immunogenetic interactions. In this study, we leveraged multiple machine learning approaches—including gradient boosting, random forest, and logistic regression—to identify key predictors of overall malignancy risk, distinguish skin versus non-skin cancers, and differentiate between skin cancer subtypes such as BCC and SCC. Our findings provide novel insights into the interplay between immunogenetic factors, clinical characteristics, and pre- and post-transplant variables, offering potential pathways for personalized risk stratification and targeted surveillance strategies in lung transplant recipients.

5.1. Predictors of de novo malignancy after lung transplantation

The success of organ transplantation depends largely on histocompatibility between donors and recipients, mediated by the human leukocyte antigen (HLA) system [44,45]. Specifically, HLA-DR alleles influence immune responses and have been linked to cancer susceptibility in transplant patients [46,47]. Our study identified HLA-DR52 as one of the most significant predictors of de novo malignancy following lung transplantation, suggesting a potential protective role. While few studies have focused on DR52 specifically, prior research shows its association with improved survival post-transplant [48] and a reduced risk of certain cancers, such as cervical cancer [49], highlighting its clinical importance in post-transplant outcomes. Furthermore, some studies have shown that HLA-DR alleles can influence responses to immunotherapy, underscoring their broader relevance in cancer prediction and treatment strategies [50].

Beyond HLA-DR52, several clinical variables were identified as significant predictors of malignancy risk. Body mass index (BMI) was associated with cancer-related factors such as chronic inflammation, hormonal imbalances, insulin resistance, and impaired immune responses [51,52]. Cytomegalovirus (CMV) IgG positivity was another strong predictor, reflecting the vulnerability of immunosuppressed patients to persistent CMV activity, which can contribute to uncontrolled inflammation and a higher risk of malignancy [53]. Elevated total serum albumin (TOT_SERUM_ALBUM) levels were also associated with increased malignancy risk, although prior studies report mixed findings on this association [54,55]. Additional influential variables included Epstein-Barr virus (EBV) serostatus, which has a well-established link to post-transplant lymphoproliferative disorder (PTLD) [56,57], and extracorporeal membrane oxygenation (ECMO) usage in lung transplant recipients, which may indirectly contribute to cancer risk due to hypoxic complications [58].

Other significant predictors included recipient age, gender, hospital length of stay (LOS), and treatment for rejection within one year (Trtrej1y). Older recipients exhibited a higher malignancy risk due to cumulative environmental exposures and immune system changes associated with aging [59]. Male recipients had an increased incidence of both skin cancers and lung cancer compared to females [60]. Prolonged hospital stays were also linked to elevated malignancy risk, potentially due to exposure to oncogenic pathogens such as human papillomavirus (HPV) and hepatitis B and C viruses [61]. Additionally, treatment for graft rejection often involves high-dose immunosuppressants, which reduce the immune system's ability to monitor and destroy cancer cells, thereby increasing susceptibility to malignancy [62,63]. Finally, donor blood urea nitrogen (BUN_DON) levels emerged as an important predictor, with elevated levels associated with poorer graft and patient survival [64] and potential links to renal cell carcinoma and breast cancer risk [65,66].

5.2. Determinants of post-transplant skin cancer risk

The gradient boosting model assessing the development of any post-transplant skin cancer identified recipient ethnicity (ETHCAT) as the strongest predictor, suggesting potential differences in genetic susceptibility, skin phototype, and environmental exposures, such as UV sensitivity, that may influence post-transplant skin cancer risk. Our analysis indicates that white recipients have a higher likelihood of developing skin cancers compared to nonwhite recipients, a finding consistent with prior research. Several studies have similarly reported that white organ transplant recipients face a significantly elevated risk of post-transplant skin cancers compared to their nonwhite counterparts [3,67,68].

Additionally, our gradient boosting model identified a significant association with recipient EBV serostatus and post-transplant skin cancer risk. Our logistics regression results confirm that EBV-positive recipients had 1.37 times higher odds of developing skin cancer compared to the reference group. Epstein-Barr Virus (EBV), a member of the herpesvirus family, is a well-established oncogenic virus known to contribute to various cancers, primarily through its ability to infect and transform cells, particularly in immunocompromised individuals [69]. While EBV is most commonly associated with lymphomas (e.g., Burkitt lymphoma, Hodgkin lymphoma) [70] and epithelial cancers (e.g., nasopharyngeal carcinoma), its potential role in skin cancer, especially in post-transplant patients, remains less direct and still under investigation [71]. Notably, EBV latent proteins, such as LMP1, mimic constitutively active CD40 signaling, activating NF-κB, JAK/STAT, and MAPK pathways. These pathways promote cell survival, proliferation, and chronic inflammation, creating a tumor-promoting microenvironment that may facilitate the development of skin cancers [72,73].

Our study found that recipient age at the time of a lung transplant is a key factor in the development of skin cancer, a finding consistent with previous research [74]. Another important variable was the most recent measured cardiac output (in liters per minute) for a transplant candidate, recorded around the time of initial transplant listing (registration). This measure, obtained as part of the hemodynamic profile through right heart catheterization (RHC), showed an inverse association with skin cancer risk, as higher cardiac output was linked to a lower likelihood of developing skin cancer following transplantation.

Our gradient boosting model identified gender as an important predictor of post-transplant skin cancer risk, with females having approximately 18% lower odds of developing skin cancer compared to males, a finding consistent with previous studies [3]. Our model also highlighted height recipients as a significant factor, showing that shorter recipients had a lower risk of developing skin cancer after lung transplantation. This aligns with prior research indicating that taller individuals generally have a slightly higher risk of several cancers, including breast, colorectal, melanoma, and prostate cancers, whereas shorter individuals tend to have a slightly lower risk [75].

Additionally, the model identified PRIOR_CARD_SURG_TCR (prior cardiac surgery at listing) as a significant variable. While several studies have investigated whether cardiac-related factors—including cardiac output, right atrial pressure, pulmonary artery pressures, and pulmonary vascular resistance—predict survival on the transplant waiting list or early post-transplant mortality, these factors have not previously been linked to cancer outcomes [76].

Furthermore, A Locus Mismatch Level (AMIS) emerged as another important variable in our study, with higher mismatch levels associated with an increased risk of developing skin cancer after lung transplantation. This finding is partially consistent with prior research but also reveals conflicting evidence. Some studies suggest that higher HLA mismatches may necessitate stronger immunosuppressive therapy, which paradoxically could maintain immune activity at a level sufficient to reduce cancer risk [77]. In contrast, other studies report that more intensive immunosuppression resulting from greater mismatches actually increases the likelihood of developing post-transplant malignancies [77].

Our gradient boosting model also identified the recipient DR1 antigen (RDR1) as a significant factor when predicting skin vs. non-skin cancers. A lower frequency of HLA-DR1 in transplant recipients was associated with a reduced risk of developing post-transplant skin cancer. This finding is consistent with prior studies; for example, Glover *et al.* (1993) reported that the presence of HLA-DR1 in renal transplant recipients was associated with a higher risk of developing non-melanoma skin cancers [78,79].

5.3. Determinants of skin cancer subtypes

When predicting basal cell carcinoma (BCC) versus squamous cell carcinoma (SCC) after lung transplantation, the random forest model demonstrated the best performance among all classifiers. The most important predictor was PRAPK_CL1 (Recipient Peak PRA% Class I @ Transplant), which measures preexisting sensitization to donor HLA Class I antigens. Our analysis suggests that recipients with low sensitization (Peak PRA Class I < 20%) at the time of transplant were more likely to develop SCC compared to BCC. This is intriguing, as lower PRA% is typically associated with a reduced rejection risk and thus less intensive immunosuppressive regimens. A potential explanation may involve immune surveillance differences: recipients with lower sensitization may have a less reactive immune repertoire, which, even under standard immunosuppression, could predispose them to more aggressive cutaneous malignancies, such as SCC.

Among lung transplant recipients, a higher number of archived donor-derived red blood cell transfusion units administered intraoperatively was associated with a lower incidence of SCC, whereas the relative incidence of BCC was higher. This finding suggests a potential cancer-type–specific association with intraoperative donor-derived transfusions. Historically, pre- or peri-transplant transfusions of donor blood were used to induce donor-specific hyporesponsiveness and improve graft acceptance. Such immunologic conditioning may also have long-term effects on immune regulation and tumor surveillance, potentially influencing the relative development of skin cancer subtypes [80].

In our cohort of post-transplant patients, lower serum albumin levels at the time of registration were significantly associated with a higher likelihood of developing cutaneous squamous cell carcinoma (SCC) compared to basal cell carcinoma (BCC). Recipients with hypoalbuminemia exhibited an increased incidence of SCC, suggesting that reduced serum albumin may serve as an independent risk factor for SCC development in the post-transplant population. As serum albumin is a well-established indicator of nutritional and systemic health [81], its reduction may reflect underlying inflammation, metabolic dysregulation, or compromised immune function [82]. These findings imply that poor nutritional or inflammatory status could exacerbate immune suppression and contribute to the heightened vulnerability to aggressive skin cancers, such as SCC, among transplant recipients.

Our random forest model identified left ventricular (LV) remodeling as an important predictor. Left ventricular (LV) remodeling describes alterations in the size, shape, and function of the heart's left ventricle that arise in response to prolonged cardiac stress. Such stress is commonly associated with chronic heart failure, myocardial infarction, valvular abnormalities, or hypertension. Importantly, LV remodeling may already be present before a patient is placed on the heart transplant waiting list [83]. In our analysis, recipients with LV remodeling showed a greater risk of developing SCC compared with BCC following lung transplantation. While LV remodeling itself has not been directly linked to skin cancer risk in transplant recipients, it is plausible that the underlying cardiovascular pathology and associated systemic inflammation could contribute to an increased susceptibility to malignancies.

Our analyses showed that the risk of developing skin cancer versus no skin cancer after lung transplantation was not significantly associated with donor insulin use. However, among recipients who developed skin cancer, those receiving organs from non-insulin-using donors had lower odds of developing SCC relative to BCC. Although few studies directly compare the effects of insulin signaling on BCC and SCC, emerging evidence highlights the role of insulin and glucose metabolism in SCC progression. For instance, Hsieh *et al.* demonstrated that systemic glucose restriction, which indirectly lowers circulating insulin levels, slows SCC progression in vivo by disrupting redox homeostasis and modulating the insulin/AKT signaling pathway [84]. This suggests that insulin signaling promotes SCC tumor growth, and reducing insulin availability—through dietary or metabolic interventions—may inhibit tumor progression. Furthermore, a recent study on head and neck squamous cell carcinoma (HNSCC) used a basal epithelial-specific GLUT1-knockout mouse model to explore insulin signaling in vivo. The researchers found that insulin receptor activity is essential for maintaining glucose uptake, redox balance, and tumor growth. Importantly, the knockdown of the insulin receptor reduced tumor proliferation and increased oxidative stress within tumors (AACR Journals, 2025). Collectively, these findings underscore the critical role of insulin signaling in SCC tumor maintenance and progression. While additional studies are needed—particularly those directly comparing SCC and BCC—current evidence strongly suggests that SCC is more dependent on insulin-related metabolic pathways *in vivo* [85].

Our model identified HLA-DR53 as being associated with an increased likelihood of developing SCC compared with BCC following lung transplantation. This finding aligns with prior studies suggesting that HLA-DR53 may influence susceptibility to SCC in patients with multiple non-melanoma skin cancers [86]. While this association does not establish direct causation, it indicates that the HLA-DR53 antigen could play a role in modulating post-transplant immune surveillance, potentially contributing to the development of specific skin cancer subtypes under chronic immunosuppression. These results highlight a possible immunogenetic factor influencing post-transplant malignancy risk and underscore the need for further investigation in clinical cohorts to confirm the association and elucidate the underlying mechanisms.

Additionally, our analysis revealed that the total number of days on the transplant waiting list was negatively correlated with the development of SCC compared to BCC. Patients who spent longer durations on the waiting list exhibited a lower incidence of SCC, suggesting that shorter wait times may be associated with an increased risk of SCC in the post-transplant population. This can be explained by the fact that patients with shorter wait times are often transplanted under more urgent or physiologically stressful conditions, which may trigger heightened systemic inflammatory responses. Both acute and chronic inflammation are known to promote DNA damage, impair immune surveillance, and enhance keratinocyte proliferation, collectively increasing SCC susceptibility. In contrast, longer waitlist durations may allow for better pre-transplant optimization and reduced inflammatory stress, thereby lowering the risk of SCC. These findings support the hypothesis that a pro-inflammatory pre-transplant environment may contribute to the preferential development of SCC over BCC in transplant recipients [87].

Our analysis also identified the recipient's hepatitis B surface antigen (HBV_SUR_ANTIGEN) status as a noteworthy variable. Although the direct association between hepatitis B surface antigen positivity and post-transplant skin cancer risk has not been extensively investigated, previous research has demonstrated a link between chronic hepatitis B virus (HBV) infection and the development of several non-hepatocellular malignancies, including cervical, breast, uterine, thyroid, lung, and cutaneous cancers. The presence of hepatitis B surface antigen indicates an active or chronic HBV infection, which can contribute to oncogenesis through mechanisms such as chronic inflammation, immune modulation, and viral integration into host DNA. These findings underscore the importance of continued cancer surveillance in HBV-positive transplant recipients, as their underlying viral infection may predispose them to malignancies beyond the liver, warranting closer post-transplant monitoring and preventive interventions [88].

Another important factor identified in our model was pan-resistant bacterial infection (RESIST_INF) at the time of registration, which indicates colonization with bacteria resistant to nearly all available antimicrobials. Although no direct link has been established between pan-resistant infection and specific post-transplant malignancies, pretransplant colonization has been shown to increase infection risk after transplant by an odds ratio of 4.7 [89]. Such infections may weaken immune function before and after transplantation, potentially contributing to immune dysregulation and increased vulnerability to malignancy.

In summary, this study demonstrates that analyzing recipient data collected before, during, and after lung transplantation using machine learning (ML) algorithms can effectively predict critical post-transplant outcomes at multiple levels: (1) The overall risk of developing de novo malignancies, (2) The likelihood of developing skin versus non-skin cancers, and (3) The risk of specific skin cancer subtypes, including BCC and SCC.

By leveraging longitudinal data from the STAR registry and employing a combination of gradient boosting, random forest, and logistic regression models, this study uncovered a range of important predictors across these three tasks. These included immunogenetic factors (e.g., HLA-DR52, HLA-DR1, HLA-DR53, and A locus mismatch), virological markers (e.g., CMV IgG and EBV serostatus), demographic variables (e.g., recipient ethnicity, age, gender, and height), and clinical indicators such as BMI, serum albumin levels, cardiac output, prior cardiac surgery, and LV remodeling. In addition, several factors were uniquely associated with skin cancer subtypes, including insulin signaling pathways, donor-derived transfusions, waiting list duration, and peak PRA Class I sensitization. Collectively, these findings provide a comprehensive framework for understanding malignancy risk following lung transplantation and highlight opportunities to develop personalized surveillance strategies.

Furthermore, this work underscores the potential of machine learning-driven screening and risk-based monitoring to inform post-transplant clinical care. By integrating complex clinical, immunogenetic, and perioperative variables, the predictive framework proposed here enables early identification of high-risk individuals, supporting targeted interventions and personalized immunosuppression strategies. Future research should focus on validating these findings across multi-center cohorts and assessing the clinical utility and cost-effectiveness of incorporating such predictive models into routine transplant care.

This study is not without limitations. First, while BMI and serum albumin (ALB) emerged as significant predictors of de novo malignancy risk, these variables can be influenced by multiple perioperative factors, including nutritional status, systemic inflammation, fluid balance, and postoperative complications. As such, their predictive value may partly reflect broader metabolic and nutritional influences rather than direct causal effects.

Second, despite leveraging a large, longitudinal dataset from the STAR registry, certain clinical details—such as primary underlying diseases, specific immunosuppressive regimens, alcohol and tobacco use—were included among the input variables but did not emerge as statistically significant predictors in the final models. Nevertheless, their potential role in post-transplant malignancy risk cannot be ruled out and warrants further investigation in prospective studies.

Third, while our models achieved robust performance, especially for overall de novo malignancy prediction and skin versus non-skin cancer classification, their generalizability may be limited by the lack of external validation. Future studies should test these predictive models on independent cohorts to strengthen their clinical applicability.

Fourth, while our machine learning models identified several important predictors, they primarily provide associative insights and do not establish causality. Integrating multi-omics data, detailed treatment protocols, and longitudinal lifestyle factors could enhance our understanding of the underlying biological mechanisms and improve prediction accuracy.

Finally, certain predictors used in our models—such as extracorporeal membrane oxygenation (ECMO_72HOURS), post-transplant rejection episodes (TRTREJ1Y), and hospital length of stay (LOS)—require observation over a defined postoperative period. These variables may introduce temporal misalignment when applied to early-stage or pre-transplant risk prediction. To assess their potential impact, we conducted a sensitivity analysis excluding these time-ambiguous variables. The results confirmed that their removal did not materially affect the ranking of key predictors (e.g., HLA-DR alleles, CMV/EBV serostatus, BMI, and serum albumin), indicating that their influence on overall model performance was minimal. Nevertheless, we acknowledge that including post-transplant variables may limit the immediate clinical applicability of the models for prospective prediction. Future work should focus on developing temporally aligned predictive frameworks. The integration of temporal or attention-based models (such as recurrent neural networks or transformer architectures) could further enhance the model's ability to capture evolving patient trajectories and support dynamic, context-aware malignancy risk predictions throughout the transplant continuum.

6. Conclusion

This study demonstrates the value of applying machine learning (ML) techniques to longitudinal lung transplant data to predict post-transplant malignancy risks at multiple levels, including overall de novo malignancies, skin versus non-skin cancers, and specific skin cancer subtypes, particularly basal cell carcinoma (BCC) and squamous cell carcinoma (SCC). By leveraging gradient boosting, random forest, and logistic regression models, we identified a set of clinically meaningful predictors that offer actionable insights for risk stratification and personalized care. Among the most important findings, HLA-DR52 was associated with a reduced likelihood of overall malignancy, whereas HLA-DR1, HLA-DR53, and A locus mismatch levels were linked to elevated risks of certain cancers, underscoring the significant role of immunogenetics in post-transplant outcomes. Recipient ethnicity, particularly white race, emerged as the strongest predictor for skin cancers, consistent with prior studies showing increased susceptibility to UV-related malignancies. Additionally, EBV serostatus, CMV IgG positivity, and BMI were key factors influencing overall cancer risk, while serum albumin levels, cardiac output, LV remodeling, and prior cardiac surgery provided additional prognostic value.

For skin cancer subtype prediction, several clinically relevant factors were identified. Peak PRA Class I sensitization, donor insulin status, and donor-derived transfusions were associated with differential risks between BCC and SCC. Furthermore, insulin signaling pathways and metabolic regulation appear to play a greater role in SCC development, highlighting potential therapeutic targets for high-risk patients. Collectively, these findings emphasize the need for tumor surveillance strategies after lung transplantation, integrating immunogenetic, clinical, and virological profiles into post-transplant care. Incorporating ML-driven predictive models into clinical workflows can enable early identification of high-risk individuals, guide tailored cancer screening, optimize immunosuppressive regimens, and facilitate timely interventions. Future research should focus on validating these predictive models across other organ transplant populations and assessing their cost-effectiveness to facilitate integration into routine clinical transplant care.

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Author contributions

Nasim Nosoudi: Conceptualization, methodology, validation, investigation, data curation, project administration, writing -– original draft. Amir Zadeh: Methodology, software, validation, formal analysis, data curation, visualization, resources, writing - original draft. Ray Nichols: Investigation, data curation, writing – original draft. Cameron Kiani: Validation, writing – review & editing. Jamie Ramirez-Vick: Supervision, resources, writing – review & editing.

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Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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